DinoFlap Runner

A little fusion between dinosaur chrome runner game and flappy bird

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Programming in Python[C]; 22-49739-3

1. Project Overview:

DinoFlap Game is a Python-based 2D game made with Pygame of the Pyhton Programming Language. It creatively fuses two well-known games _ the Chrome Dinosaur Runner and Flappy Bird into one unified experience. The game supports two separate control modes chosen by the player before the game begins: Keyboard Mode and Camera Mode.

2. Track & External Library:

Item	Detail
Track	Track 1 — Game (Pygame)
External Libraries	pygame, opencv-python, mediapipe, numpy
Language	Python
Storage	JSON (data/scores)

3. Setup & Environment Settings:

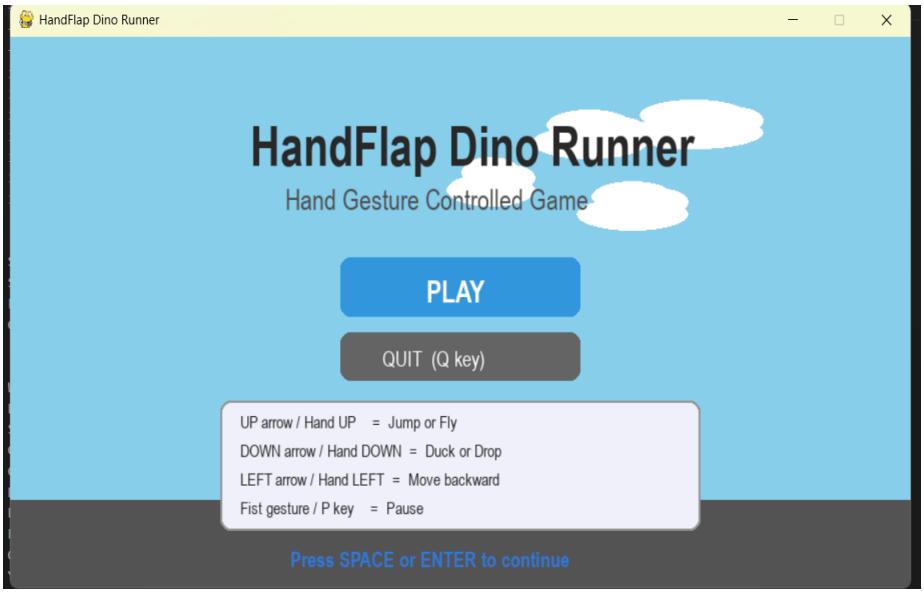
1. **Create virtual environment:** python -m venv .venv
2. **Activate (Windows):** .venv\Scripts\activate
3. **Install libraries:** pip install -r requirements.txt
4. **Run the game:** python game.py
5. **Freeze after changes:** pip freeze > requirements.txt

Requirements.txt:

```
pygame>=2.1.0
opencv-python>=4.5.0
mediapipe>=0.10.30
numpy>=1.21.0
```

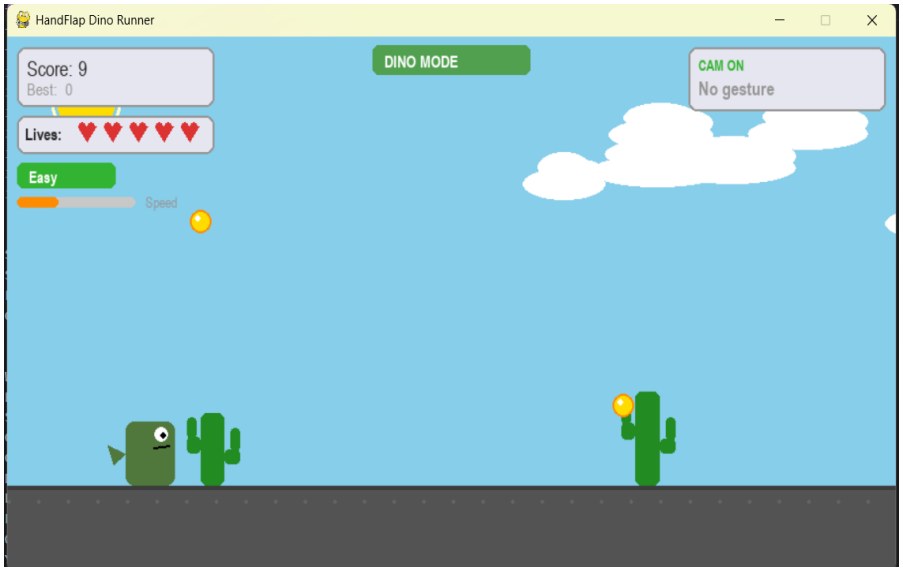
4. Features:

Gameplay:

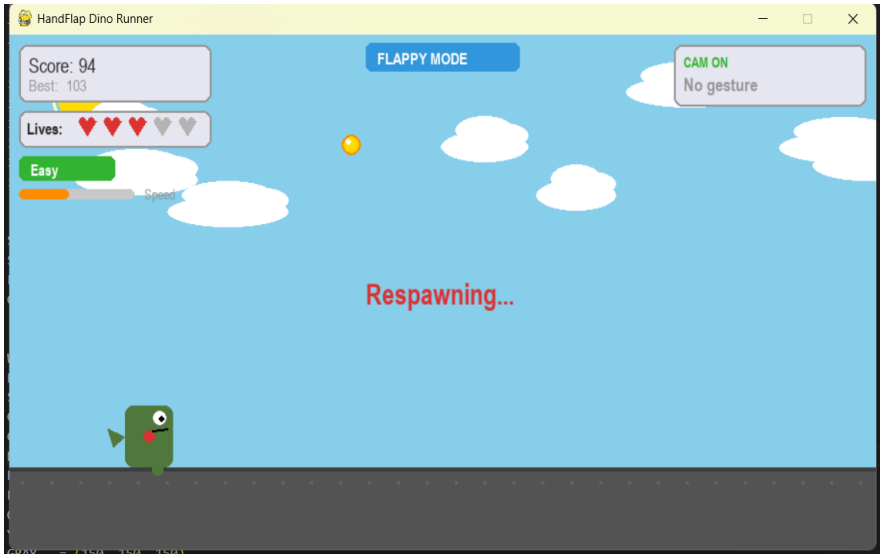


Player should select the play button by pressing Space bar or Enter

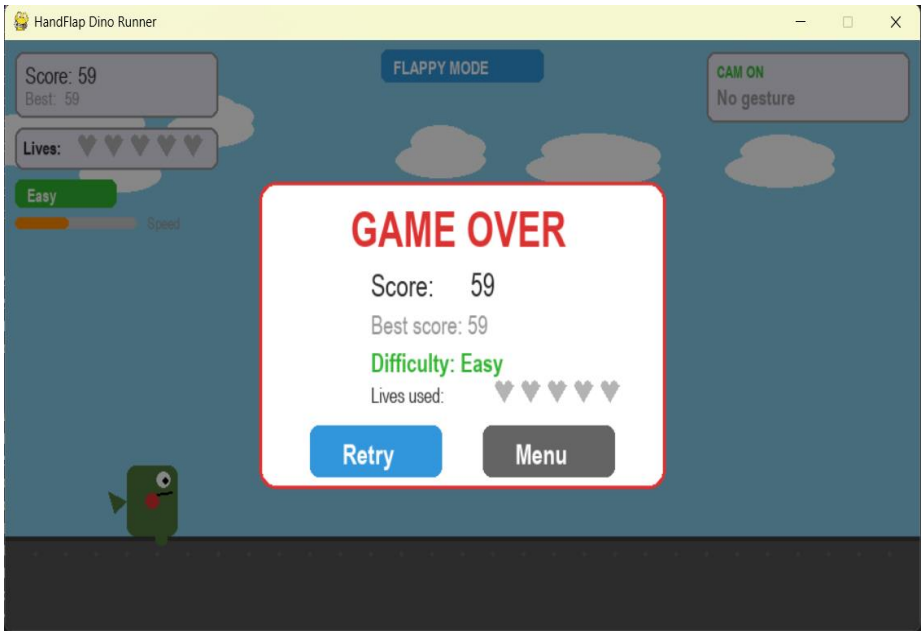
Three difficulty cards: Easy (green), Medium (orange), Hard (red). Each card shows the difficulty name, heart icons for lives, and a description. The selected mode badge appears at the top center.



Dinosaur running on the ground approaching a cactus obstacle. Gold coin are floating mid-air. Scores are showing and lives as hearts.



Dinosaur flying through a gap between pipes.



Game over in red, final score, best score for this difficulty and difficulty name in its color. Empty heart icons for lives used and retry/ menu buttons